An Introduction to DAHA Involution and Macdonald Duality

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July 30, 2026

Main Statements

Statement (Root-system and DAHA involution)**.** *For the dual affine root system R^ι,*

$$\alpha_j^\iota = \frac{A_j}{\sqrt{p}}, \qquad \mathring{Q}^\iota = \frac{1}{\sqrt{p}} M, \qquad M^\iota = \sqrt{p}\, \mathring{Q}. \tag{1}$$

There is an involution

$$\phi_{\widetilde{W}} : \mathcal{H}_{\widetilde{W}} \longrightarrow \mathcal{H}_{\widetilde{W}^\iota} \tag{2}$$

given by

$$\phi_{\widetilde{W}}(T_j) = (T_j^\iota)^{-1}, \qquad \phi_{\widetilde{W}}(Y_\mu) = X^\iota_{\mu/\sqrt{p}}, \qquad \phi_{\widetilde{W}}(X_\beta) = Y^\iota_{\sqrt{p}\,\beta}, \tag{3}$$

and

$$\phi_{\widetilde{W}}(q) = (q^\iota)^{-1}, \qquad \phi_{\widetilde{W}}(t_j^{1/2}) = (t_j^\iota)^{-1/2}. \tag{4}$$

For $B_n^{(1)}$, we have $(B_n^{(1)})^\iota = C_n^{(1)}$ and $p = 2$.

Statement (Non-symmetric Macdonald polynomials)**.** *For generic parameters and $\lambda \in \mathring{Q}$, there is a unique monic triangular common Y-eigenvector*

$$E_\lambda^R = X_\lambda + \sum_{\eta \prec \lambda} c^R_{\lambda\eta} X_\eta, \qquad Y_\mu E_\lambda^R = \gamma_\lambda^R(\mu) E_\lambda^R \quad (\mu \in M). \tag{5}$$

Statement (Non-symmetric evaluation-duality)**.** *Define*

$$X^\iota_{\mu/\sqrt{p}}\big(\mathrm{sp}_R(\lambda)\big) = \gamma_\lambda^R(-\mu), \qquad X_\lambda\big(\mathrm{sp}_{R^\iota}(\beta)\big) = \gamma_\beta^{R^\iota}(-\sqrt{p}\,\lambda). \tag{6}$$

Then

$$\boxed{\frac{E_\lambda^R\big(\mathrm{sp}_{R^\iota}(\beta)\big)}{E_\lambda^R\big(\mathrm{sp}_{R^\iota}(0)\big)} = \frac{E_\beta^{R^\iota}\big(\mathrm{sp}_R(\lambda)\big)}{E_\beta^{R^\iota}\big(\mathrm{sp}_R(0)\big)}}. \tag{7}$$

The details and the $B_n^{(1)}$–$C_n^{(1)}$ calculation follow.

Contents

1 Notation and Conventions

We use Ion's conventions. Thus $X_\delta = q^{-1}$, the X-lattice is $\mathring{Q}$, and the Y-lattice is M. Operators act from right to left.

Symbol	Meaning
$R,\ R^\iota$	An affine root system and its dual.
$\alpha_0, \ldots, \alpha_n$	The affine simple roots; $\alpha_1, \ldots, \alpha_n$ are the finite simple roots.
$\delta,\ \theta$	The null root and the highest finite root; $\alpha_0 = \delta - \theta$.
$\mathring{Q}$	The finite root lattice and the index lattice for X.
M	The translation lattice generated by $A_j = e_j \alpha_j$, and the index lattice for Y.
$\mathring{W},\ W$	The finite and affine Weyl groups.
λ_μ	Translation by $\mu \in M$.
$X_\beta,\ Y_\mu$	The two commuting lattice directions in DAHA.
T_j	The Hecke generator for s_j.
$p = e_s$	The scaling number; $p = 2$ in types $B_n^{(1)}$ and $C_n^{(1)}$.
$\phi_{\widetilde{W}}$	The DAHA Involution.
$\gamma_\lambda^R(\mu)$	The Y_μ-eigenvalue of E_λ^R.
$\mathrm{sp}_R(\lambda)$	The spectral character attached to λ.
E_λ^R	The non-symmetric Macdonald polynomial.

2 Root Data

Let

$$A = (a_{ij})_{0 \le i,j \le n} \tag{8}$$

be an irreducible affine Cartan matrix, with finite part

$$\mathring{A} = (a_{ij})_{1 \le i,j \le n}. \tag{9}$$

Let

$$\alpha_0, \alpha_1, \ldots, \alpha_n \tag{10}$$

be the affine simple roots, and set

$$\mathring{Q} = \bigoplus_{j=1}^{n} \mathbb{Z}\alpha_j. \tag{11}$$

Write

$$\delta = \sum_{j=0}^{n} a_j \alpha_j, \qquad \delta^\vee = \sum_{j=0}^{n} a_j^\vee \alpha_j^\vee. \tag{12}$$

We exclude type $A^{(2)}_{2n}$. Then $a_0 = 1$ and

$$\alpha_0 = \delta - \theta, \tag{13}$$

where θ is the highest finite root.

For $1 \le j \le n$, set

$$d_j := \frac{a_j}{a_j^\vee}, \qquad e_j := \max\{d_j, a_0^{-1}\}, \qquad A_j := e_j \alpha_j, \tag{14}$$

and

$$M := \bigoplus_{j=1}^{n} \mathbb{Z} A_j. \tag{15}$$

Write λ_μ for translation by $\mu \in M$. The value of e_j depends only on the root length. Let e_s and e_l be its values on short and long roots, and set

$$p := e_s. \tag{16}$$

In the simply laced case, $p = 1$.

Use the symmetric form

$$(\alpha_j, \alpha_k) := d_j^{-1} a_{jk}, \qquad 0 \le j, k \le n, \tag{17}$$

and the reflections

$$s_\alpha(x) = x - \frac{2(x, \alpha)}{(\alpha, \alpha)} \alpha. \tag{18}$$

The finite Weyl group has the presentation

$$\mathring{W} = \left\langle s_i \ (1 \le i \le n) \ \middle| \ s_i^2 = 1, \ \underbrace{s_i s_j s_i \cdots}_{m_{ij} \text{ factors}} = \underbrace{s_j s_i s_j \cdots}_{m_{ij} \text{ factors}} (i \ne j) \right\rangle, \tag{19}$$

where

$$m_{ij} = \begin{cases} 2, & a_{ij} a_{ji} = 0, \\ 3, & a_{ij} a_{ji} = 1, \\ 4, & a_{ij} a_{ji} = 2, \\ 6, & a_{ij} a_{ji} = 3, \\ \infty, & a_{ij} a_{ji} \ge 4. \end{cases} \tag{20}$$

See [Kac83, §3.13]. The affine Weyl group is

$$W = \mathring{W} \ltimes M. \tag{21}$$

3 The Double Affine Hecke Algebra

Let m be the least common denominator of

$$\{(\alpha_j, \Lambda_k) \mid 1 \le j, k \le n\}, \tag{22}$$

where Λ_k is the k-th fundamental weight. Let $\mathbb{F}$ be the field of rational functions in $q^{1/m}$ and the $t_j^{1/2}$.

The two lattice algebras are

$$\mathbb{F}[M] = \bigoplus_{\mu \in M} \mathbb{F} Y_\mu, \qquad \mathbb{F}[\mathring{Q}] = \bigoplus_{\beta \in \mathring{Q}} \mathbb{F} X_\beta, \tag{23}$$

with

$$Y_\mu Y_\nu = Y_{\mu+\nu}, \qquad X_\beta X_\gamma = X_{\beta+\gamma}. \tag{24}$$

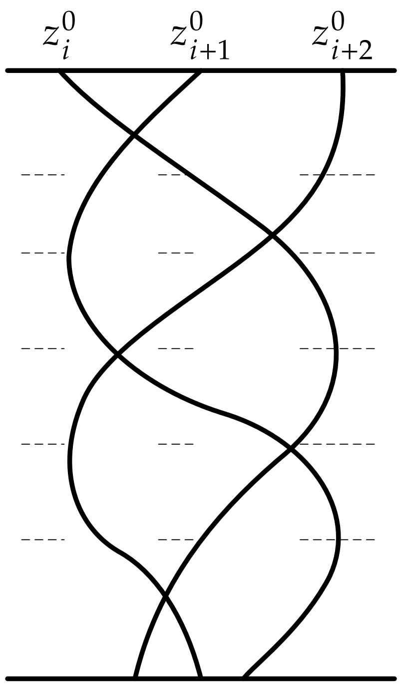
The parameters depend only on root length:

$$d_j = d_k \quad \Longleftrightarrow \quad t_j = t_k. \tag{25}$$

3.1 The finite Hecke algebra

Since we omitted the reflection relations on may want to know what is the inverse of T_j. The finite Hecke algebra $\mathcal{H}_{\mathring{W}}$ is generated by $T_1, \ldots, T_n$. It has the braid relations from (19) and the quadratic relations

$$T_j - T_j^{-1} = t_j^{1/2} - t_j^{-1/2}. \tag{26}$$



$$\sigma_i \, \sigma_{i+1} \, \sigma_i \, \sigma_{i+1}^{-1} \, \sigma_i^{-1} \, \sigma_{i+1}^{-1}$$
$$= \text{identity}$$

3.2 The affine Hecke algebra

The affine Hecke algebra $\mathcal{H}_W$ is generated by $\mathcal{H}_{\mathring{W}}$ and $\mathbb{F}[M]$, with

$$Y_\mu T_j - T_j Y_{s_j(\mu)} = \left(t_j^{1/2} - t_j^{-1/2}\right) \frac{Y_\mu - Y_{s_j(\mu)}}{1 - Y_{A_j}}, \tag{27}$$

for $\mu \in M$ and $1 \le j \le n$.

3.3 The double affine Hecke algebra

Definition 3.1. *The double affine Hecke algebra $\mathcal{H}_{\widetilde{W}}$ is generated by $\mathcal{H}_W$ and $\mathbb{F}[\mathring{Q}]$, with*

$$T_j X_\beta - X_{s_j(\beta)} T_j = \left(t_j^{1/2} - t_j^{-1/2}\right) \frac{X_\beta - X_{s_j(\beta)}}{1 - X_{-\alpha_j}}, \tag{28}$$

for $\beta \in \mathring{Q}$ and $0 \le j \le n$, together with

$$X_\delta = q^{-1}. \tag{29}$$

For $j = 0$, use $X_\delta = q^{-1}$ to read $X_{s_0(\beta)}$.

Thus DAHA contains the two commuting lattice algebras

$$\mathbb{F}[\mathring{Q}] \quad \text{generated by the } X_\beta, \tag{30}$$

and

$$\mathbb{F}[M] \quad \text{generated by the } Y_\mu. \tag{31}$$

4 The Root-System and DAHA Involutions

4.1 The root-system involution

Definition 4.1. *For $1 \le j \le n$, define*

$$\alpha_j^\iota := \frac{A_j}{\sqrt{p}} = \frac{e_j}{\sqrt{p}} \alpha_j, \tag{32}$$

and set

$$\delta^\iota = \delta. \tag{33}$$

These data define the affine root system R^ι. Its affine simple root α_0^ι is fixed by its affine type.

Under the natural identification of the finite root spaces,

$$\mathring{Q}^\iota = \frac{1}{\sqrt{p}} M, \qquad M^\iota = \sqrt{p}\, \mathring{Q}. \tag{34}$$

The simply laced types are fixed. In particular,

$$A_n^{(1)\,\iota} = A_n^{(1)}. \tag{35}$$

For the classical non-simply-laced types,

$$B_n^{(1)\,\iota} = C_n^{(1)}, \qquad C_n^{(1)\,\iota} = B_n^{(1)}. \tag{36}$$

The map ι swaps short and long finite roots.

4.2 The DAHA isomorphism

Write the generators of $\mathcal{H}_{\widetilde{W}^\iota}$ as

$$T_j^\iota, \qquad X_\gamma^\iota, \qquad Y_\nu^\iota. \tag{37}$$

Theorem 4.2 (Ion's DAHA involution)**.** *The assignments*

$$\phi_{\widetilde{W}}(T_j) = (T_j^\iota)^{-1}, \qquad 1 \le j \le n, \tag{38}$$

$$\phi_{\widetilde{W}}(Y_\mu) = X^\iota_{\mu/\sqrt{p}}, \qquad \mu \in M, \tag{39}$$

$$\phi_{\widetilde{W}}(X_\beta) = Y^\iota_{\sqrt{p}\,\beta}, \qquad \beta \in \mathring{Q}, \tag{40}$$

extend to an involution

$$\phi_{\widetilde{W}} : \mathcal{H}_{\widetilde{W}} \longrightarrow \mathcal{H}_{\widetilde{W}^\iota}, \tag{41}$$

with

$$\phi_{\widetilde{W}}(q) = (q^\iota)^{-1}, \qquad \phi_{\widetilde{W}}(t_j^{1/2}) = (t_j^\iota)^{-1/2}. \tag{42}$$

Its inverse is $\phi_{\widetilde{W}^\iota}$.

This is [Ion03, Theorems 2.2–2.3]. In short,

$$\boxed{X_\beta \longleftrightarrow Y^\iota_{\sqrt{p}\,\beta}}. \tag{43}$$

5 Non-symmetric Macdonald Polynomials

The polynomial representation is

$$\mathcal{P}_R := \mathbb{F}[\mathring{Q}]. \tag{44}$$

For $\lambda \in \mathring{Q}$, let λ_+ be the dominant element of $\mathring{W}\lambda$. Let $v(\lambda)$ be the shortest element of $\mathring{W}$ that sends λ to the anti-dominant chamber. Define

$$\lambda \succeq \eta \quad \Longleftrightarrow \quad \begin{cases} \lambda_+ > \eta_+, & \text{in strict dominance order,} \\ \lambda_+ = \eta_+ \text{ and } v(\lambda) \le v(\eta), & \text{in finite Bruhat order.} \end{cases} \tag{45}$$

For dominant elements $\xi, \zeta \in \mathring{Q}$, the dominance order is defined by

$$\xi \le_{\text{dom}} \zeta \quad \Longleftrightarrow \quad \xi - \zeta \in \sum_{i=1}^n \mathbb{Z}_{\ge 0}\alpha_i. \tag{46}$$

Set

$$\mathring{Q}_{\preceq\lambda} := \{\eta \in \mathring{Q} \mid \eta \preceq \lambda\}, \qquad \mathcal{P}_{\preceq\lambda} := \operatorname{span}_{\mathbb{F}}\{X_\eta \mid \eta \preceq \lambda\}. \tag{47}$$

The set $\mathring{Q}_{\preceq\lambda}$ is finite and lower closed.

Theorem 5.1 (Y-triangularity). *For $\mu \in M$ and $\lambda \in \mathring{Q}$, there are unique coefficients $\gamma_\lambda^R(\mu) \in \mathbb{F}^\times$ and $a_{\eta\lambda}^R(\mu) \in \mathbb{F}$ such that*

$$Y_\mu X_\lambda = \gamma_\lambda^R(\mu) X_\lambda + \sum_{\eta \prec \lambda} a_{\eta\lambda}^R(\mu) X_\eta. \tag{48}$$

Hence every Y_μ preserves $\mathcal{P}_{\preceq\lambda}$, see [Mac03, 4.6.4–4.6.12].

Definition 5.2. *For generic parameters, $E_\lambda^R \in \mathcal{P}_R$ is the monic common eigenvector defined by*

$$E_\lambda^R = X_\lambda + \sum_{\eta \prec \lambda} c_{\lambda\eta}^R X_\eta, \tag{49}$$

$$Y_\mu E_\lambda^R = \gamma_\lambda^R(\mu) E_\lambda^R, \qquad \mu \in M. \tag{50}$$

Here generic means that the required eigenvalues are distinct and the denominators below are nonzero.

Proposition 5.3 (Triangular recursion). *Choose $\nu \in M$ such that the eigenvalues $\gamma_\eta^R(\nu)$, $\eta \preceq \lambda$, are pairwise distinct.*

Fix $\eta \prec \lambda$. Suppose that all coefficients $c_{\lambda\zeta}^R$ with $\eta \prec \zeta \prec \lambda$ are already known. Define

$$E_\lambda^{(>\eta)} := X_\lambda + \sum_{\eta \prec \zeta \prec \lambda} c_{\lambda\zeta}^R X_\zeta. \tag{51}$$

This is the part of E_λ^R already computed above η. It is only a partial sum, not a new Macdonald polynomial.

Then

$$\boxed{c_{\lambda\eta}^R = \frac{[X_\eta]\, Y_\nu E_\lambda^{(>\eta)}}{\gamma_\lambda^R(\nu) - \gamma_\eta^R(\nu)}.} \tag{52}$$

Thus the coefficients are computed from top to bottom.

Example 5.4. *Suppose*

$$\lambda \succ \zeta \succ \eta \tag{53}$$

and

$$Y_\nu X_\lambda = \gamma_\lambda X_\lambda + a_{\zeta\lambda} X_\zeta + a_{\eta\lambda} X_\eta + \cdots, \tag{54}$$

$$Y_\nu X_\zeta = \gamma_\zeta X_\zeta + a_{\eta\zeta} X_\eta + \cdots. \tag{55}$$

First,

$$c_{\lambda\zeta} = \frac{a_{\zeta\lambda}}{\gamma_\lambda - \gamma_\zeta}. \tag{56}$$

When computing $c_{\lambda\eta}$, the partial sum is

$$E_\lambda^{(>\eta)} = X_\lambda + c_{\lambda\zeta} X_\zeta. \tag{57}$$

Therefore

$$[X_\eta]\, Y_\nu E_\lambda^{(>\eta)} = a_{\eta\lambda} + c_{\lambda\zeta} a_{\eta\zeta}, \tag{58}$$

and hence

$$c_{\lambda\eta} = \frac{a_{\eta\lambda} + c_{\lambda\zeta} a_{\eta\zeta}}{\gamma_\lambda - \gamma_\eta}. \tag{59}$$

Since $T_i 1 = t_i^{1/2} 1$, every Y_μ acts on 1 by a scalar. Therefore

$$\boxed{E_0^R = 1.} \tag{60}$$

6 Macdonald Duality

6.1 Spectral points

For $\lambda \in \mathring{Q}$, define the character

$$\mathrm{sp}_R(\lambda) : \mathcal{P}_{R^\iota} \longrightarrow \mathbb{F}, \qquad \boxed{X^\iota_{\mu/\sqrt{p}}(\mathrm{sp}_R(\lambda)) := \gamma_\lambda^R(-\mu)}, \quad \mu \in M. \tag{61}$$

This defines it on all of $\mathcal{P}_{R^\iota} = \mathbb{F}[\mathring{Q}^\iota]$, since $\mathring{Q}^\iota = p^{-1/2}M$. For $\beta \in \mathring{Q}^\iota$, define

$$\mathrm{sp}_{R^\iota}(\beta) : \mathcal{P}_R \longrightarrow \mathbb{F}, \qquad \boxed{X_\lambda(\mathrm{sp}_{R^\iota}(\beta)) := \gamma_\beta^{R^\iota}(-\sqrt{p}\,\lambda)}, \quad \lambda \in \mathring{Q}. \tag{62}$$

6.2 The duality theorem

Theorem 6.1. *[Mac03, 4.7.14] For $\lambda \in \mathring{Q}$ and $\beta \in \mathring{Q}^\iota$,*

$$\boxed{\frac{E_\lambda^R(\mathrm{sp}_{R^\iota}(\beta))}{E_\lambda^R(\mathrm{sp}_{R^\iota}(0))} = \frac{E_\beta^{R^\iota}(\mathrm{sp}_R(\lambda))}{E_\beta^{R^\iota}(\mathrm{sp}_R(0))}.} \tag{63}$$

7 Examples

7.1 Parameter conventions for the B_n–C_n calculation

Write

$$q_B, \quad t_l^B, \quad t_s^B \tag{64}$$

for the parameters of $B_n^{(1)}$, and

$$q_C, \quad t_s^C, \quad t_l^C \tag{65}$$

for those of $C_n^{(1)}$. The subscripts always refer to root length in the indicated root system.

Remark 7.1 (Parameter convention in duality). *The map ϕ_{BC} produces the $C_n^{(1)}$-realization with inverted parameters*

$$\phi_{BC}(q_B) = q_C^{-1}, \qquad \phi_{BC}(t_l^B) = (t_s^C)^{-1}, \qquad \phi_{BC}(t_s^B) = (t_l^C)^{-1}, \tag{66}$$

whereas $E_\beta^{C_n^{(1)}}$ is written in the standard $C_n^{(1)}$-parameters. Thus we need to introduce

$$\overline{q_C} = q_C^{-1}, \qquad \overline{t_s^C} = (t_s^C)^{-1}, \qquad \overline{t_l^C} = (t_l^C)^{-1}. \tag{67}$$

Then the coefficient identification used in evaluation-duality is

$$\jmath_{BC} := {}^{-} \circ \phi_{BC}, \tag{68}$$

so that

$$\boxed{\jmath_{BC}(q_B) = q_C, \qquad \jmath_{BC}(t_l^B) = t_s^C, \qquad \jmath_{BC}(t_s^B) = t_l^C.} \tag{69}$$

7.2 The root-system involution

The finite simple roots of types A, B, and C are

$$\begin{aligned}
A_{n-1}: &\quad \alpha_i = \varepsilon_i - \varepsilon_{i+1}, && 1 \le i < n; && \\
B_n: &\quad \alpha_i = \varepsilon_i - \varepsilon_{i+1}, && 1 \le i < n, && \alpha_n = \varepsilon_n; \\
C_n: &\quad \alpha_i = \varepsilon_i - \varepsilon_{i+1}, && 1 \le i < n, && \alpha_n = 2\varepsilon_n.
\end{aligned}$$

Table 1: Affine labels and dual labels.

Type	$(a_0, a_1, \ldots, a_n)$	$(a_0^\vee, a_1^\vee, \ldots, a_n^\vee)$
$A_n^{(1)}$	$(1, 1, \ldots, 1)$	$(1, 1, \ldots, 1)$
$B_n^{(1)}$	$(1, 1, 2, \ldots, 2, 2)$	$(1, 1, 2, \ldots, 2, 1)$
$C_n^{(1)}$	$(1, 2, \ldots, 2, 1)$	$(1, 1, \ldots, 1)$

Hence

$$A_n^{(1)}: \quad M = \mathring{Q}, \tag{70}$$

$$B_n^{(1)}: \quad M = \bigoplus_{i=1}^{n-1} \mathbb{Z}\alpha_i \oplus \mathbb{Z}(2\alpha_n), \tag{71}$$

$$C_n^{(1)}: \quad M = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i) \oplus \mathbb{Z}\alpha_n. \tag{72}$$

Now take $B_n^{(1)}$. Then

$$d_1 = \cdots = d_{n-1} = 1, \qquad d_n = 2, \tag{73}$$

so

$$e_1 = \cdots = e_{n-1} = 1, \qquad e_n = 2. \tag{74}$$

Thus

$$A_j = \alpha_j \quad (1 \le j < n), \qquad A_n = 2\alpha_n, \qquad p = e_s = 2. \tag{75}$$

Therefore

$$\alpha_i^\iota = \frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \quad (i < n), \qquad \alpha_n^\iota = \sqrt{2}\,\varepsilon_n. \tag{76}$$

Their squared lengths are

$$(\alpha_i^\iota, \alpha_i^\iota) = 1 \quad (i < n), \qquad (\alpha_n^\iota, \alpha_n^\iota) = 2. \tag{77}$$

The first $n-1$ roots are short and the last is long. The finite Cartan matrix is

$$\mathring{A}^\iota = \begin{pmatrix} 2 & -1 & 0 & \cdots & 0 \\ -1 & 2 & -1 & \ddots & \vdots \\ 0 & -1 & \ddots & \ddots & 0 \\ \vdots & \ddots & \ddots & 2 & -2 \\ 0 & \cdots & 0 & -1 & 2 \end{pmatrix}, \tag{78}$$

so

$$\boxed{(B_n^{(1)})^\iota = C_n^{(1)}.} \tag{79}$$

In this case, Theorem 4.2 reads

$$\begin{aligned}
\phi_{\widetilde{W}}(T_j) &= (T_j^\iota)^{-1}, && 1 \le j \le n, && (80)\\
\phi_{\widetilde{W}}(Y_\mu) &= X^\iota_{\mu/\sqrt{2}}, && \mu \in M, && (81)\\
\phi_{\widetilde{W}}(X_\beta) &= Y^\iota_{\sqrt{2}\beta}, && \beta \in \mathring{Q}, && (82)\\
\phi_{\widetilde{W}}(q) &= (q^\iota)^{-1}, && && (83)\\
\phi_{\widetilde{W}}(t_j) &= (t_j^\iota)^{-1}. && && (84)
\end{aligned}$$

Also,

$$\mathring{Q}^\iota = \frac{1}{\sqrt{2}} M = \bigoplus_{i=1}^{n-1} \mathbb{Z}\frac{\varepsilon_i - \varepsilon_{i+1}}{\sqrt{2}} \oplus \mathbb{Z}(\sqrt{2}\varepsilon_n), \tag{85}$$

and

$$M^\iota = \bigoplus_{i=1}^{n-1} \mathbb{Z}(2\alpha_i^\iota) \oplus \mathbb{Z}\alpha_n^\iota = \sqrt{2}\,\mathring{Q}. \tag{86}$$

Particularly, we have

$$\boxed{\phi_{\widetilde{W}}\left((t_l^B)^{1/2}\right) = (t_s^C)^{-1/2}, \qquad \phi_{\widetilde{W}}\left((t_s^B)^{1/2}\right) = (t_l^C)^{-1/2}.} \tag{87}$$

Thus the X-lattice of B_n is the Y-lattice of C_n, and the Y-lattice of B_n is the X-lattice of C_n.

7.3 The polynomial $E_{\varepsilon_1}^{B_n^{(1)}}$

Lemma 7.2 (Triangular support of ε_1). *For the order in (45),*

$$\{\eta \in \mathring{Q} : \eta \preceq \varepsilon_1\} = \{\varepsilon_1, 0\}. \tag{88}$$

Consequently,

$$\mathcal{P}_{\preceq \varepsilon_1} = \mathrm{span}_{\mathbb{F}}\{x_1, 1\}. \tag{89}$$

Proof. Write

$$\eta_+ = \sum_{j=1}^{n} m_j \varepsilon_j. \tag{90}$$

For type B_n,

$$\alpha_i^\vee = \varepsilon_i - \varepsilon_{i+1} \quad (i < n), \qquad \alpha_n^\vee = 2\varepsilon_n. \tag{91}$$

Hence

$$(\eta_+, \alpha_i^\vee) = m_i - m_{i+1} \quad (i < n), \qquad (\eta_+, \alpha_n^\vee) = 2m_n. \tag{92}$$

Therefore η_+ is dominant if and only if

$$m_1 \geq m_2 \geq \cdots \geq m_n \geq 0. \tag{93}$$

Suppose that $\eta_+ \leq_{\mathrm{dom}} \varepsilon_1$. Then

$$\varepsilon_1 - \eta_+ = \sum_{i=1}^{n} c_i \alpha_i, \qquad c_i \geq 0. \tag{94}$$

Comparing coordinates gives

$$1 - m_1 = c_1, \qquad -m_j = c_j - c_{j-1} \quad (2 \leq j \leq n). \tag{95}$$

If $\eta_+ \neq 0$, then $m_1 \geq 1$. Thus $m_1 = 1$ and $c_1 = 0$. The remaining equations, together with $m_j, c_j \geq 0$, give successively

$$m_j = c_j = 0 \qquad (2 \leq j \leq n). \tag{96}$$

Hence η_+ is either 0 or ε_1.

It remains to consider

$$\mathring{W}\varepsilon_1 = \{\pm\varepsilon_i : 1 \leq i \leq n\}. \tag{97}$$

For the shortest element sending a weight to $-\varepsilon_1$,

$$\ell(v(\varepsilon_i)) = 2n - i, \qquad \ell(v(-\varepsilon_i)) = i - 1. \tag{98}$$

Thus $v(\varepsilon_1)$ is the unique element of maximal length. Since Bruhat order is length-increasing,

$$v(\varepsilon_1) \leq v(\eta) \tag{99}$$

forces $\eta = \varepsilon_1$. $\square$

It follows that

$$E_{\varepsilon_1}^{B_n^{(1)}} = x_1 + c_B. \tag{100}$$

The operators in this formula carry their $B_n^{(1)}$-parameters. where

$$t_0^B = t_1^B = \cdots = t_{n-1}^B = t_l^B, \qquad t_n^B = t_s^B. \tag{101}$$

For $1 \le i < n$, s_i swaps x_i and x_{i+1}. The reflection s_n sends x_n to x_n^{-1}, and

$$(s_0 f)(x_1, x_2, x_3, \dots, x_n) = f\left(\frac{1}{qx_2}, \frac{1}{qx_1}, x_3, \dots, x_n\right). \tag{102}$$

Set

$$L_B := (t_l^B)^{1/2}, \qquad S_B := (t_s^B)^{1/2}. \tag{103}$$

Since $X_{-\alpha_0} = qx_1x_2$,

$$T_i f = L_B s_i f + (L_B - L_B^{-1})\frac{f - s_i f}{1 - x_{i+1}/x_i}, \qquad 1 \le i < n, \tag{104}$$

$$T_n f = S_B s_n f + (S_B - S_B^{-1})\frac{f - s_n f}{1 - x_n^{-1}}, \tag{105}$$

$$T_0 f = L_B s_0 f + (L_B - L_B^{-1})\frac{f - s_0 f}{1 - qx_1x_2}. \tag{106}$$

For $\theta = \varepsilon_1 + \varepsilon_2$, let

$$R_2^B = T_2T_3 \cdots T_{n-1}T_nT_{n-1} \cdots T_3T_2. \tag{107}$$

By Ion's result ,

$$Y_{-\theta}^{B_n^{(1)}} = R_2^B T_1 R_2^B T_0. \tag{108}$$

From

$$s_0 x_1 = q_B^{-1} x_2^{-1}, \qquad \frac{x_1 - q_B^{-1}x_2^{-1}}{1 - q_B x_1 x_2} = -q_B^{-1}x_2^{-1}, \tag{109}$$

we obtain

$$T_0 x_1 = q_B^{-1} L_B^{-1} x_2^{-1}. \tag{110}$$

Applying (108) from right to left gives

$$\boxed{Y_{-\theta}^{B_n^{(1)}} x_1 = q_B^{-1}(t_l^B)^{-1}x_1 - q_B^{-1}(t_l^B)^{n-2}(t_s^B - 1).} \tag{111}$$

Since the word in (108) contains $4n-6$ long-root generators and two short-root generators,

$$\boxed{Y_{-\theta}^{B_n^{(1)}} 1 = (t_l^B)^{2n-3} t_s^B.} \tag{112}$$

Set

$$a_B := q_B^{-1}(t_l^B)^{-1}, \tag{113}$$

$$b_B := -q_B^{-1}(t_l^B)^{n-2}(t_s^B - 1), \tag{114}$$

$$d_B := (t_l^B)^{2n-3} t_s^B. \tag{115}$$

Then

$$\gamma_{\varepsilon_1}^{B_n^{(1)}}(-\theta) = a_B, \qquad \gamma_0^{B_n^{(1)}}(-\theta) = d_B. \tag{116}$$

If $a_B \neq d_B$, the eigenvalue equation for (100) gives

$$\boxed{c_B = \frac{q_B^{-1}(t_l^B)^{n-2}(t_s^B - 1)}{(t_l^B)^{2n-3}t_s^B - q_B^{-1}(t_l^B)^{-1}}.} \tag{117}$$

Therefore

$$\boxed{E_{\varepsilon_1}^{B_n^{(1)}} = x_1 + \frac{q_B^{-1}(t_l^B)^{n-2}(t_s^B - 1)}{(t_l^B)^{2n-3}t_s^B - q_B^{-1}(t_l^B)^{-1}}.} \tag{118}$$

The space $\mathcal{P}_{\leq \varepsilon_1}$ is invariant under every Y_μ. Since the a_B-eigenspace of $Y_{-\theta}^{B_n^{(1)}}$ on this space is one-dimensional, the polynomial in (118) is a common Y-eigenvector.

7.4 The corresponding polynomial of type $C_n^{(1)}$

Set

$$\eta_i := \frac{\varepsilon_i}{\sqrt{2}}, \qquad \beta_0 := \eta_1 + \eta_2 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}, \qquad y_i := X_{\eta_i}^{\iota}. \tag{119}$$

The variables y_i belong to an ambient Laurent polynomial ring; all monomials below lie in $\mathbb{F}[\mathring{Q}^{\iota}]$.

Lemma 7.3 (Triangular support of β_0). *For the $C_n^{(1)}$-order,*

$$\{\beta \in \mathring{Q}^{\iota} : \beta \preceq \beta_0\} = \{\beta_0, 0\}. \tag{120}$$

Proof. In the η-coordinates,

$$\mathring{Q}^{\iota} = \left\{ \sum_{j=1}^{n} m_j \eta_j : m_j \in \mathbb{Z}, \ \sum_{j=1}^{n} m_j \equiv 0 \pmod 2 \right\}. \tag{121}$$

Write

$$\beta_+ = \sum_{j=1}^{n} m_j \eta_j. \tag{122}$$

Since

$$(\eta_i, \eta_j) = \frac{1}{2}\delta_{ij}, \tag{123}$$

the simple coroots of the scaled C_n-system are

$$(\alpha_i^{\iota})^{\vee} = 2(\eta_i - \eta_{i+1}) \quad (i < n), \qquad (\alpha_n^{\iota})^{\vee} = 2\eta_n. \tag{124}$$

Consequently,

$$(\beta_+, (\alpha_i^{\iota})^{\vee}) = m_i - m_{i+1} \quad (i < n), \qquad (\beta_+, (\alpha_n^{\iota})^{\vee}) = m_n. \tag{125}$$

Thus β_+ is dominant if and only if

$$m_1 \geq m_2 \geq \cdots \geq m_n \geq 0. \tag{126}$$

Suppose that

$$\beta_+ \leq_{\mathrm{dom}} \beta_0, \qquad \beta_0 = \eta_1 + \eta_2. \tag{127}$$

Then

$$\beta_0 - \beta_+ = \sum_{i=1}^{n} c_i \alpha_i^\iota, \qquad c_i \geq 0. \tag{128}$$

The sum of the η-coordinates of the right-hand side is $2c_n$. Hence

$$\sum_{i=1}^{n} m_i = 2 - 2c_n \leq 2. \tag{129}$$

This sum is nonnegative and even. If it is nonzero, the decreasing nonnegative coordinates m_i are therefore either

$$(1,1,0,\ldots,0) \quad \text{or} \quad (2,0,\ldots,0). \tag{130}$$

The second possibility is excluded because

$$\beta_0 - 2\eta_1 = \eta_2 - \eta_1 = -\alpha_1^\iota \tag{131}$$

is not a nonnegative combination of simple roots. Thus

$$\beta_+ = 0 \quad \text{or} \quad \beta_+ = \beta_0. \tag{132}$$

Finally, let

$$J = \{i : s_i\beta_0 = \beta_0\}. \tag{133}$$

If w_J is the longest element of W_J, then

$$v(\beta_0) = w_0 w_J \tag{134}$$

is the unique maximal minimal representative of $\mathring{W}/W_J$. The elements $v(\beta)$, with $\beta \in \mathring{W}\beta_0$, run through these representatives. Therefore

$$v(\beta_0) \leq v(\beta) \tag{135}$$

forces $\beta = \beta_0$. □

Hence

$$E_{\beta_0}^{C_n^{(1)}} = X_{\beta_0}^\iota + c_C. \tag{136}$$

Put

$$L_C := (t_s^C)^{1/2}, \qquad S_C := (t_l^C)^{1/2}. \tag{137}$$

Here L_C is the short-root parameter and S_C is the long-root parameter. Since $-2\eta_1 \in M^\iota$ is anti-dominant,

$$Y_{-2\eta_1}^{C_n^{(1)}} = T_1 T_2 \cdots T_{n-1} T_n T_{n-1} \cdots T_2 T_1 T_0. \tag{138}$$

In particular, $T_1, \ldots, T_{n-1}$ carry L_C, whereas T_0, T_n carry S_C.

Since

$$s_0(y_1) = q_C^{-1} y_1^{-1}, \qquad X^\iota_{-\alpha_0^\iota} = q_C y_1^2, \tag{139}$$

we have

$$T_0(y_1 y_2) = q_C^{-1} S_C^{-1} y_1^{-1} y_2. \tag{140}$$

Applying the remaining operators in (138) gives

$$\boxed{Y^{C_n^{(1)}}_{-2\eta_1} X^\iota_{\beta_0} = q_C^{-1}(t_s^C)^{-(n-2)}(t_l^C)^{-1} X^\iota_{\beta_0} + q_C^{-1}(t_s^C)^{n-2}(1 - t_s^C).} \tag{141}$$

The word in (138) contains $2n-2$ short-root generators and two long-root generators. Therefore

$$\boxed{Y^{C_n^{(1)}}_{-2\eta_1} 1 = (t_s^C)^{n-1} t_l^C.} \tag{142}$$

Set

$$a_C := q_C^{-1}(t_s^C)^{-(n-2)}(t_l^C)^{-1}, \tag{143}$$

$$b_C := q_C^{-1}(t_s^C)^{n-2}(1 - t_s^C), \tag{144}$$

$$d_C := (t_s^C)^{n-1} t_l^C. \tag{145}$$

Then

$$\gamma^{C_n^{(1)}}_{\beta_0}(-2\eta_1) = a_C, \qquad \gamma^{C_n^{(1)}}_{0}(-2\eta_1) = d_C. \tag{146}$$

If $a_C \neq d_C$, the eigenvalue equation yields

$$\boxed{c_C = \frac{q_C^{-1}(t_s^C)^{n-2}(t_s^C - 1)}{(t_s^C)^{n-1} t_l^C - q_C^{-1}(t_s^C)^{-(n-2)}(t_l^C)^{-1}}.} \tag{147}$$

Thus

$$\boxed{E^{C_n^{(1)}}_{\beta_0} = X^\iota_{\beta_0} + \frac{q_C^{-1}(t_s^C)^{n-2}(t_s^C - 1)}{(t_s^C)^{n-1} t_l^C - q_C^{-1}(t_s^C)^{-(n-2)}(t_l^C)^{-1}}.} \tag{148}$$

7.5 An explicit B_n–C_n duality identity

Take

$$\lambda = \varepsilon_1, \qquad \beta_0 = \frac{\varepsilon_1 + \varepsilon_2}{\sqrt{2}}. \tag{149}$$

The spectral-point definitions give

$$X^\iota_{\beta_0}\big(\mathrm{sp}_B(\varepsilon_1)\big) = a_B, \qquad X^\iota_{\beta_0}\big(\mathrm{sp}_B(0)\big) = d_B, \tag{150}$$

$$x_1\big(\mathrm{sp}_C(\beta_0)\big) = a_C, \qquad x_1\big(\mathrm{sp}_C(0)\big) = d_C. \tag{151}$$

Transporting the $B_n^{(1)}$-coefficients to the $C_n^{(1)}$-coefficient field by $\jmath_{BC}$, the normalized duality identity is Therefore

$$\frac{E^R_\lambda\big(\mathrm{sp}_{R^\iota}(\beta_0)\big)}{E^R_\lambda\big(\mathrm{sp}_{R^\iota}(0)\big)} = \frac{a_C + \jmath_{BC}(c_B)}{d_C + \jmath_{BC}(c_B)}, \tag{152}$$

and

$$\frac{E_{\beta_0}^{R^\iota}(\mathrm{sp}_R(\lambda))}{E_{\beta_0}^{R^\iota}(\mathrm{sp}_R(0))} = \frac{\jmath_{BC}(a_B) + c_C}{\jmath_{BC}(d_B) + c_C}. \tag{153}$$

Thus the duality becomes

$$\boxed{\frac{a_C + \jmath_{BC}(c_B)}{d_C + \jmath_{BC}(c_B)} = \frac{\jmath_{BC}(a_B) + c_C}{\jmath_{BC}(d_B) + c_C}.} \tag{154}$$

Equivalently, use the common notation

$$q := q_B = q_C, \qquad t_l := t_l^B = t_s^C, \qquad t_s := t_s^B = t_l^C. \tag{155}$$

Then

$$a_B = q^{-1} t_l^{-1}, \qquad d_B = t_l^{2n-3} t_s, \tag{156}$$

$$a_C = q^{-1} t_l^{-(n-2)} t_s^{-1}, \qquad d_C = t_l^{n-1} t_s, \tag{157}$$

and

$$c_B = \frac{q^{-1} t_l^{n-2}(t_s - 1)}{t_l^{2n-3} t_s - q^{-1} t_l^{-1}}, \tag{158}$$

$$c_C = \frac{q^{-1} t_l^{n-2}(t_l - 1)}{t_l^{n-1} t_s - q^{-1} t_l^{-(n-2)} t_s^{-1}}. \tag{159}$$

Thus (154) becomes

$$\boxed{\frac{a_C + c_B}{d_C + c_B} = \frac{a_B + c_C}{d_B + c_C}.} \tag{160}$$

Both sides are equal to

$$\boxed{\frac{q^{-1} t_l^{n-2}(t_l + t_s - 1) - q^{-2} t_l^{-(n-1)} t_s^{-1}}{t_l^{3n-4} t_s^2 - q^{-1} t_l^{n-2}}.} \tag{161}$$

This verifies evaluation-duality for $\lambda = \varepsilon_1$ and $\beta_0 = (\varepsilon_1 + \varepsilon_2)/\sqrt{2}$.

8 Guiding Principle

The X-operators are coordinates:

$$X_\beta \quad \text{acts by multiplication.} \tag{162}$$

The Y-operators are spectral operators:

$$Y_\mu \quad \text{acts by } q\text{-difference operators.} \tag{163}$$

The DAHA involution swaps them:

$$\boxed{\text{coordinates} \longleftrightarrow \text{spectral variables.}} \tag{164}$$

This gives Macdonald duality.

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